Detecting Meaning Drift in Twitter Sentiment Streams: Informed Ageing Naive Bayes with an Error-Based ADWIN Detector

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Abstract—Sentiment classifiers deployed on social media streams degrade when the stream changes, a phenomenon known as concept drift. Iosifidis et al. adapt an ageing Multinomial Naive Bayes (MNB) classifier when a vocabulary-based detector signals change. Such a detector, however, only sees changes in which words are used; it cannot see a change in what they mean. We reproduce the informed ageing MNB on the Sentiment140 stream and add an ADWIN detector on the prequential error stream, fused with the vocabulary detector. The system runs end to end on Apache Kafka and Spark Structured Streaming with metrics in InfluxDB and live dashboards in Grafana. On controlled drift scenarios with known drift points, ADWIN detects a label flip after 39 instances instead of 3,999, recovers in 3,183 instances instead of 8,204 and raises accuracy from 71.0% to 73.8% (5 seeds). When only one topic's meaning changes, the vocabulary detector misses the drift in 2 of 5 runs while ADWIN detects all of them. On the real streams the enhancement improves accuracy from 80.8% to 81.3–81.7% (DS1) and from 83.0% to 83.7–84.1% (DS2), adding about 10 ms of model time per 1,000 tweets in the streaming pipeline.

Index Terms—concept drift, data streams, sentiment analysis, ADWIN, naive Bayes, Apache Kafka, Spark Structured Streaming

I. Introduction

Opinions on social media form an unbounded, non-stationary stream: topics come and go, vocabulary evolves and real-world events shift the sentiment towards an entity. Bechini et al. show that a stance classifier trained once and deployed on Twitter loses accuracy as events change the discussion, and that adaptive learning schemes recover it [1]. In streaming terms this is concept drift: the joint distribution $P(X, y)$ changes over time [2]. A change in $P(X)$ alone (new words, new topics) is virtual drift; a change in $P(y \mid X)$ (the same words now signal a different class) is real drift, and only real drift necessarily hurts a classifier.

Iosifidis et al. propose informed adaptation for sentiment streams [3]: an ageing MNB classifier forgets old observations through an ageing factor λ, and a vocabulary-based change detector decides when to forget faster or rebuild. The detector compares the vocabulary of recent tweets with that of past tweets, so it reacts to virtual drift by construction. The authors note that their detector reacts with delay because vocabularies of different topics overlap [3]. A drift in meaning with an unchanged vocabulary, such as a word that changes polarity after an event, is invisible to it.

This project asks whether monitoring the classifier's own errors closes that gap. Our contributions are:

- A reproduction of the informed ageing MNB with all five λ strategies on the Sentiment140 stream and a volatile variant, including fixes to the detector that the paper leaves underspecified (Section III).
- An enhancement that adds ADWIN [4] on the prequential error stream, with four fusion modes and a rebuild reaction on the post-drift window.
- Controlled drift scenarios with known drift points, following Costa et al. [5], including a label-flip scenario that isolates real drift with an unchanged vocabulary, and an evaluation of detection delay, false alarms, misses, recovery time and cost over 224 runs.
- An end-to-end streaming system on Kafka, Spark, InfluxDB and Grafana that runs both models side by side on identical data.

II. Related Work

Concept drift. Gama et al. survey drift detection and adaptation and distinguish blind adaptation (forgetting at a fixed rate, e.g. a sliding or damped window) from informed adaptation that reacts to detected change [2]. ADWIN keeps an adaptive window over a numeric stream and cuts it when two sub-windows have significantly different means, with rigorous false-positive and false-negative bounds [4]; applied to a 0/1 error stream it is a standard error-rate drift detector.

Sentiment streams. Bifet and Frank study sentiment classification on Twitter as a stream and its evaluation [6]. Wagner et al. introduce the ageing MNB, which weights class and word counts by the recency of their last observation [7]. Iosifidis et al. make the ageing factor adaptive with a vocabulary-based detector and five tuning strategies, evaluated on Sentiment140 [3], [8].

Drift types on Twitter. Costa et al. simulate sudden, gradual, incremental and recurring drift by retimestamping real tweets and show that no single learning strategy is best for all drift types [5]. Bechini et al. document event-driven drift in a real Twitter stance stream [1]. We borrow the scenario methodology of the former and the motivation of the latter.

Systems. Kafka provides an ordered, replayable log for event streams [9]; Spark Structured Streaming processes it in micro-batches [10]; River provides online learning and drift detectors in Python [11].

III. Method

A. Ageing Multinomial Naive Bayes

Following [3], [7], the model stores for every class c its count N_c and last observation time t^c_{lo}, and for every (word, class) pair its count N_{wc} and last time $t^{w,c}_{lo}$. At time t counts are aged on read:

$$P^t(c) \propto N_c \, e^{-\lambda(t-t^c_{lo})}, \tag{1}$$

$$P^t(w \mid c) = \frac{N_{wc} \, e^{-\lambda(t-t^{w,c}_{lo})} + \alpha}{\sum_j N_{jc} \, e^{-\lambda(t-t^{j,c}_{lo})} + \alpha|V|}, \tag{2}$$

with Laplace smoothing $\alpha = 1$; scores are computed in log space and unseen words are ignored. With $\lambda = 0$ the model is the accumulative MNB. The sum in the denominator of (2) runs over the whole vocabulary; we cache it per class relative to a reference time, replace a word's term in $O(1)$ when it is learnt and rescale the cache before $e^{\lambda(t-t_{ref})}$ can overflow, so prediction costs $O(|d|)$ per tweet. Unit tests check the cached form against the literal equations and the accumulative model against scikit-learn's MultinomialNB.

Time unit. We measure t in hours from the tweet timestamp. With the paper's $\lambda = 0.2$ per instance, a word unseen for 100 tweets keeps $e^{-20} \approx 2 \cdot 10^{-9}$ of its weight; on DS1 this drops pre-change accuracy to 68.0%, while per hour the model behaves as reported in the paper.

B. Vocabulary-Based Change Detector (Baseline)

Every w instances the detector compares, per class, the vocabulary V_{sl} of the latest window with the vocabulary V of a reference window by precision $= |V_{sl} \cap V|/|V|$ and flags a change if the precision leaves $\mu \pm \alpha_d \sigma$ ($\alpha_d = 1.8$) and a warning at $\beta\sigma$ ($\beta = 0.334$), where μ, σ are the moving mean and standard deviation of past checks [3]. Two details are not fixed by the paper and matter in practice. An accumulated reference vocabulary grows while V_{sl} does not, so precision falls steadily (0.45 to 0.10 over 80k stable tweets) and the detector fires constantly; we use the previous window as reference, which makes the statistic stationary. Moving statistics over 10 checks produced many false alarms; 20 checks remove them on our sudden-drift scenario.

On a change, one of five strategies adapts the model [3]: SlowIncreaseUpToALimit (λ += c up to λ_{max}), SlowIncreaseFastReset (the same, reset after one window at λ_{max}), FastSetFastReset ($\lambda = \lambda_{max}$ for one window), FastSetSlowDecrease ($\lambda = \lambda_{max}$, then -5% per window) and Rebuild (a new model on the instances buffered since the warning). Each comes in a -Zero- ($\lambda_0 = 0$) and an -Init- ($\lambda_0 > 0$) variant.

C. Enhancement: ADWIN on the Prediction Error

Prequential evaluation tests each tweet before learning it, so the model observes the error $e_i = [\hat{y}_i \neq y_i]$ for free. We feed e_i to ADWIN [4], [11] with $\delta = 0.01$ and count only cuts after which the error rate went up. Because ADWIN watches $P(y \mid X)$ through the errors, it sees real drift regardless of the vocabulary.

The two detectors drive two channels (Algorithm 1): vocabulary signals go to the λ strategy as in the baseline, ADWIN signals trigger a rebuild on ADWIN's post-drift window (its current width, at least 100 instances). The fusion mode decides how they combine: vocab_only (the baseline), adwin_only, or (both channels act; our main model) and and (a signal counts only if the other detector fired within w instances). Each channel has its own cooldown of 2,000 instances, and ADWIN starts after a 1,000-instance warm-up.

Three design choices came from failures we observed. (i) Resetting ADWIN after every rebuild blinded it: a vocabulary false alarm one instance before the drift left a fresh ADWIN without a record of the normal error level. ADWIN is therefore never reset; it drops its old window itself when it fires. (ii) A shared cooldown let that same false alarm swallow ADWIN's detection 64 instances later, hence per-channel cooldowns. (iii) A minimum rebuild size of 1,000 padded the rebuild with old-concept tweets because ADWIN fires within tens of instances; a minimum of 100 reduced recovery on the label flip from 5,626 to 3,023 instances.

D. Streaming Architecture

Fig. 1 shows the system. A producer replays a stream in timestamp order into a single-partition Kafka topic (one partition preserves order). Spark executors parse the JSON and tokenise each tweet with the same function as the offline runner. Ageing MNB is sequential and stateful, so each micro-batch is collected to the driver inside foreachBatch, where the baseline and the enhanced model process it side by side, one tweet at a time in Kafka offset order. Per-model metrics, pipeline metrics and drift events go to InfluxDB and appear in Grafana with drift annotations. All models, detectors and metrics live in one Python package shared by the streaming consumer and an offline runner that replays the same parquet streams without Kafka for the experiment grid.

IV. Experimental Setup

A. Streams

DS1 is Sentiment140 [8]: 1.6M tweets from 6 April to 25 June 2009 with emoticon-derived labels. The file is

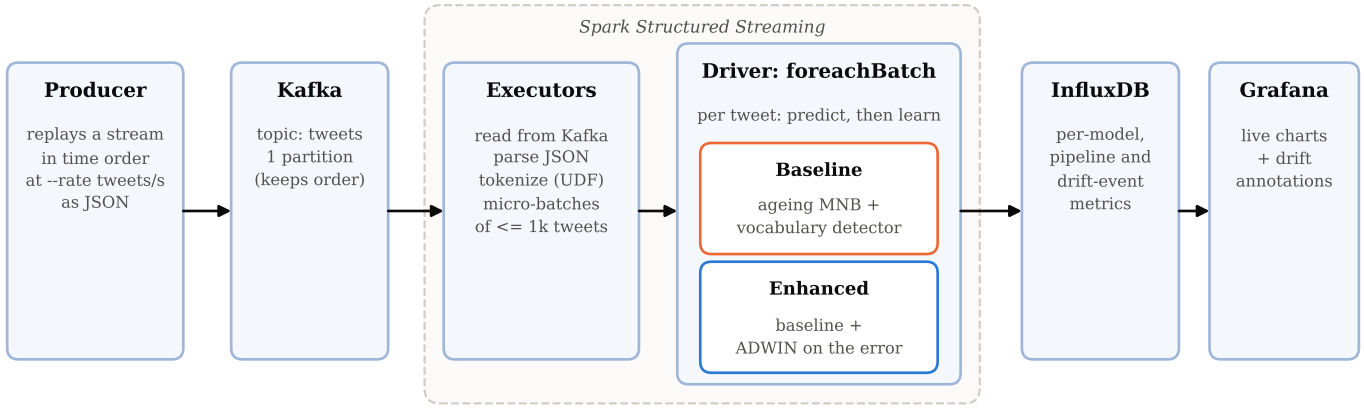


Fig. 1. System architecture. Spark executors tokenise; the driver runs both models on identical micro-batches and writes metrics and drift events to InfluxDB for Grafana.

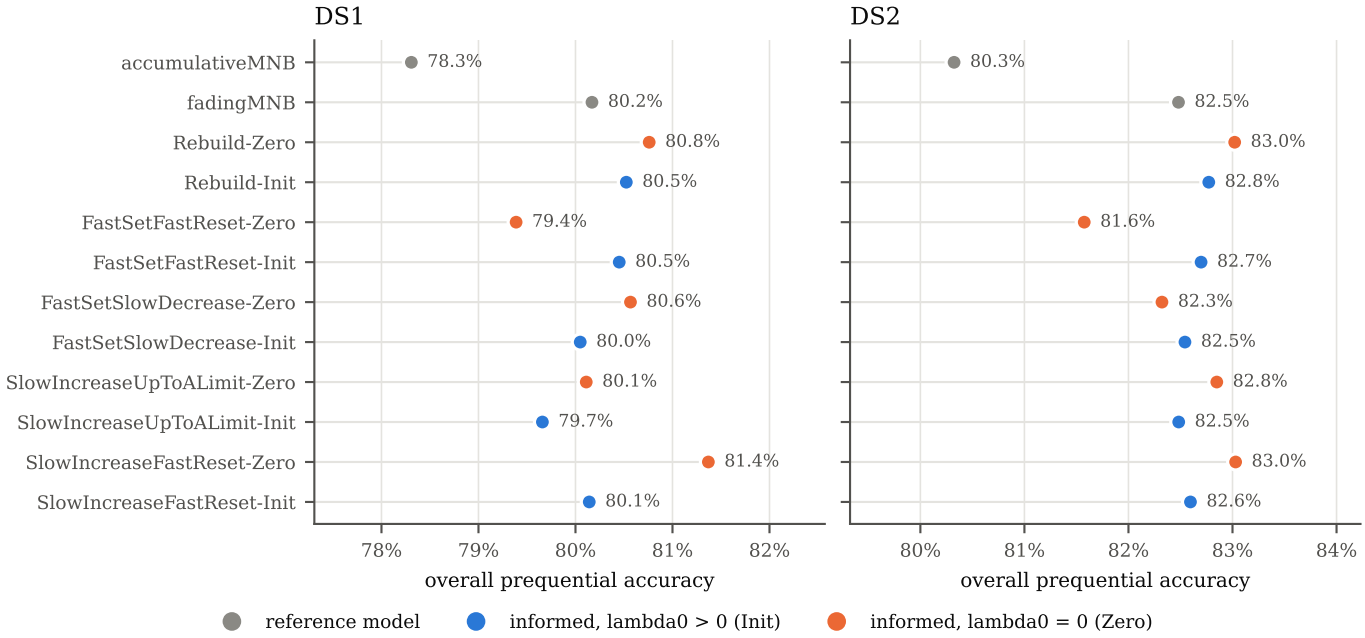


Fig. 2. Baseline reproduction: overall accuracy of the reference models and all informed strategies on DS1 and DS2 (Table 1 parameters of [3]).

sorted by label, so we sort by timestamp (Fig. 3). The last positive tweet is instance 1,324,774; from 1,324,775 the stream is all negative, matching the paper's change point at about 1,326,000. 1,685 tweet ids occur twice with opposite labels (label noise); we keep them in DS1 and DS2. DS2 is the paper's volatile variant: removing class fractions from six equal segments of the pre-change stream gives exactly the paper's 1,073,065 tweets (378,288 positive), with the positive share swinging 68/27/70/17/56/28% and then 0%. The paper does not specify which instances it removed, so the segment pattern is our own.

Controlled scenarios re-timestamp real tweets (one per second, 80,000 per stream) following Costa et al. [5], with keyword-defined topics A (work/school) and B (music/movies), each balanced 50/50 by class and without the duplicated ids: sudden (A to B at 40k), gradual (B's share rises from 0 to 1 over 30k–50k), recurring (A, B, A, B every 20k), prior shift (50% to 10% positive at 40k), label flip (every label inverts at 40k; $P(X)$ and $P(y)$ unchanged) and label flip (topic) (only topic A's labels invert). Each is generated with 5 seeds.

B. Models and Parameters

Reference models are the accumulative MNB and fading MNB (fixed λ; 0.2 on DS1 and the scenarios, 0.3 on DS2). The baseline reproduction uses Table 1 of [3] ($w = 24$k on DS1, 22k on DS2; λ_{max} for the SlowIncrease strategies, missing from the table, set to 0.5). In the grid we keep

TABLE I
Prequential accuracy (%) per stream; scenarios: mean ± std over 5 seeds.

Model	DS1	DS2	label flip	label flip (topic)	prior shift	sudden	gradual	recurring
accumulative MNB	78.3	80.3	53.3 ± 0.1	62.3 ± 0.1	80.3 ± 0.1	73.9 ± 0.1	73.8 ± 0.1	74.0 ± 0.1
fading MNB	80.2	82.5	55.7 ± 0.1	62.3 ± 0.1	80.1 ± 0.1	73.8 ± 0.1	73.8 ± 0.1	73.9 ± 0.1
baseline: Rebuild-Zero	80.8	83.0	71.0 ± 1.0	63.1 ± 0.4	81.6 ± 0.3	73.9 ± 0.2	73.7 ± 0.1	73.3 ± 0.2
baseline: SlowIncreaseFastReset-Zero	81.4	83.0	54.5 ± 0.6	62.2 ± 0.1	80.2 ± 0.1	73.7 ± 0.1	73.7 ± 0.1	73.8 ± 0.1
enhanced: ADWIN only	81.6	84.1	74.3 ± 0.1	64.0 ± 0.2	80.3 ± 0.1	73.9 ± 0.1	73.8 ± 0.1	73.7 ± 0.3
enhanced: OR	81.3	83.7	73.8 ± 0.3	63.5 ± 0.5	81.6 ± 0.3	73.9 ± 0.2	73.7 ± 0.1	73.2 ± 0.1
enhanced: AND	81.7	84.0	73.4 ± 0.9	63.8 ± 0.5	80.3 ± 0.1	73.9 ± 0.1	73.8 ± 0.1	73.6 ± 0.3

Algorithm 1 Enhanced informed ageing MNB, one prequential step

Require: tweet (d, y, t), model M, vocabulary detector D_V, ADWIN D_A, strategy S

$\hat{y} \leftarrow M.\text{predict}(d, t)$; $M.\text{learn}(d, y, t)$
append (d, y, t) to the recent buffer B
if $i \geq$ warm-up and $D_A.\text{update}([\hat{y} \neq y])$ signals a rise then
  if fusion $\neq$ and or D_V fired within w then
    if ADWIN channel not cooling down then
      $M \leftarrow$ new model trained on the last $\max(D_A.\text{width}, 100)$ items of B
    end if
  end if
end if
if fusion $\neq$ adwin_only and $D_V.\text{add}(d, y)$ completes a window then
  $s \leftarrow$ signal of D_V ▷ none / warning / change
  if fusion = and then $s \leftarrow$ change iff $s \neq$ none and D_A fired within w
  end if
  apply $S.\text{onCheck}(s)$ to M ▷ set λ or rebuild
end if

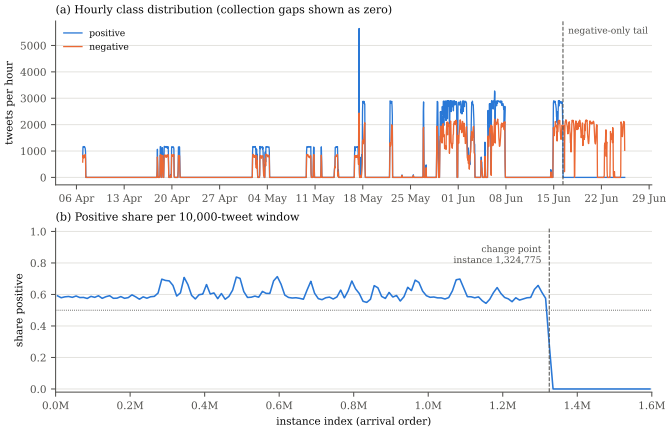


Fig. 3. DS1: (a) hourly class counts (cf. Fig. 1a of [3]); (b) positive share per 10k tweets with the natural change point.

the two best baselines on DS1/DS2, Rebuild-Zero and SlowIncreaseFastReset-Zero, and build the enhancement on Rebuild-Zero so that the only difference from that baseline is the drift signal. Scenarios use $w = 4{,}000$. ADWIN's $\delta = 0.01$ and $\alpha_d = 1.8$ were chosen by a sweep of $\delta \in \{0.002, 0.01, 0.05\}$ and $\alpha_d \in \{1.5, 1.8, 2.5\}$ over all scenarios and 3 seeds (432 runs): every enhanced setting beat every baseline setting, and $\delta = 0.01$ gave the fewest false alarms at equal accuracy.

C. Metrics

All models are evaluated prequentially (test, then train). We report accuracy and macro precision, recall and F1. A drift at p (gradual: spanning $[p, e)$) counts as detected if a detection the model acted on falls into $[p, e + T)$ with $T = 10$k for the scenarios and 50k for DS1/DS2; the first such detection gives the delay, detections outside every interval are false alarms. Recovery time is the offset after p of the first window (1k, DS: 10k instances) whose accuracy reaches 95% of the pre-drift level, searched up to the next drift. Cost is throughput, processing time per 1k-tweet batch and model size in (word, class) entries offline, and latency and model time per tweet in Spark. DS1/DS2 ground truth contains only the designed drift points, so their false-alarm counts are upper bounds.

V. Results

A. Baseline Reproduction

Fig. 2 reproduces the paper's Fig. 2. The ordering holds on both streams: the accumulative MNB is worst (78.3% on DS1, 80.3% on DS2), fixed ageing helps (80.2%, 82.5%) and informed strategies reach up to 81.4% and 83.0%. After DS1's change point the adaptive models reach 93–99.9% while the accumulative MNB stays at 77%. Two differences from the paper: our accumulative baseline is about 5 points stronger (paper: about 73%), and -Zero- variants beat -Init- ones, because constant ageing costs 1–2 points in stable periods. A sweep of the fading λ on DS1 (0.02–0.2 per hour) shows no value that avoids this trade-off (pre-change accuracy 77.8–76.3%, tail 92–99%), which supports the paper's own argument for forgetting only on detected change.

B. The Enhancement on Real Drift

Table I lists accuracy per stream and Table II the main results. On the label flip (Fig. 6) every model collapses to about 25% when the labels invert. The accumulative

TABLE II

Main results over the grid (7 models x 8 streams, scenarios x 5 seeds). Acc., P, R, F1 in %; macro P/R/F1 averaged over all runs; detection over all 47 true drifts (FA = false alarms); delay and recovery in instances (recovery averages the drifts a model recovers from; Never rec. counts the others); cost from the offline runner (throughput, p95 time per 1k-tweet batch, (word, class) entries).

Model	Acc. scen.	Acc. DS1	Acc. DS2	P	R	F1	Detected	FA	Delay	Recovery	Never rec.	Tweets/s	Batch p95 (ms)	Entries
accumulative MNB	69.6	78.3	80.3	69.6	69.7	69.6	-	-	-	1,685	10	78k	22	75k
fading MNB	69.9	80.2	82.5	70.0	70.1	70.0	-	-	-	310	10	46k	35	75k
baseline: Rebuild-Zero	72.8	80.8	83.0	72.9	72.4	72.6	36/47	70	4,766	2,230	6	38k	55	42k
baseline: SlowIncreaseFastReset-Zero	69.7	81.4	83.0	69.8	69.9	69.9	36/47	70	4,766	906	10	43k	37	75k
enhanced: ADWIN only	73.3	81.6	84.1	73.3	73.4	73.3	18/47	29	839	821	6	46k	33	54k
enhanced: OR	73.3	81.3	83.7	73.4	72.9	73.1	39/47	102	2,290	1,247	6	37k	58	40k
enhanced: AND	73.1	81.7	84.0	73.1	73.2	73.1	18/47	30	1,837	1,165	6	41k	37	54k

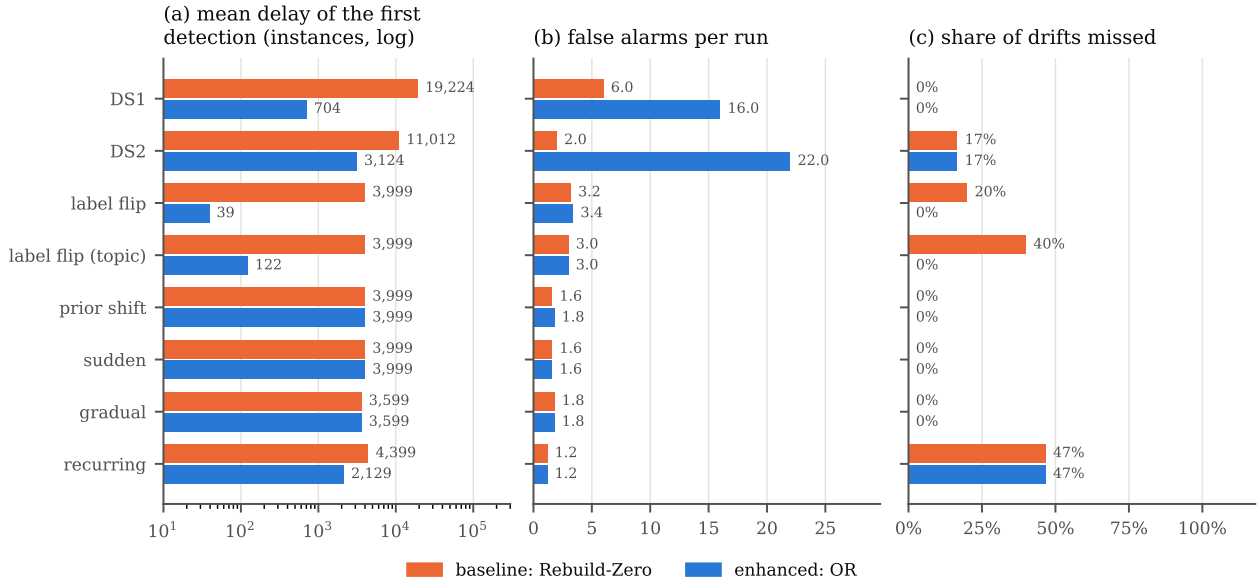


Fig. 4. Detection per stream for the baseline and the enhanced (OR) model: (a) mean delay of the first detection, (b) false alarms per run, (c) share of drifts missed.

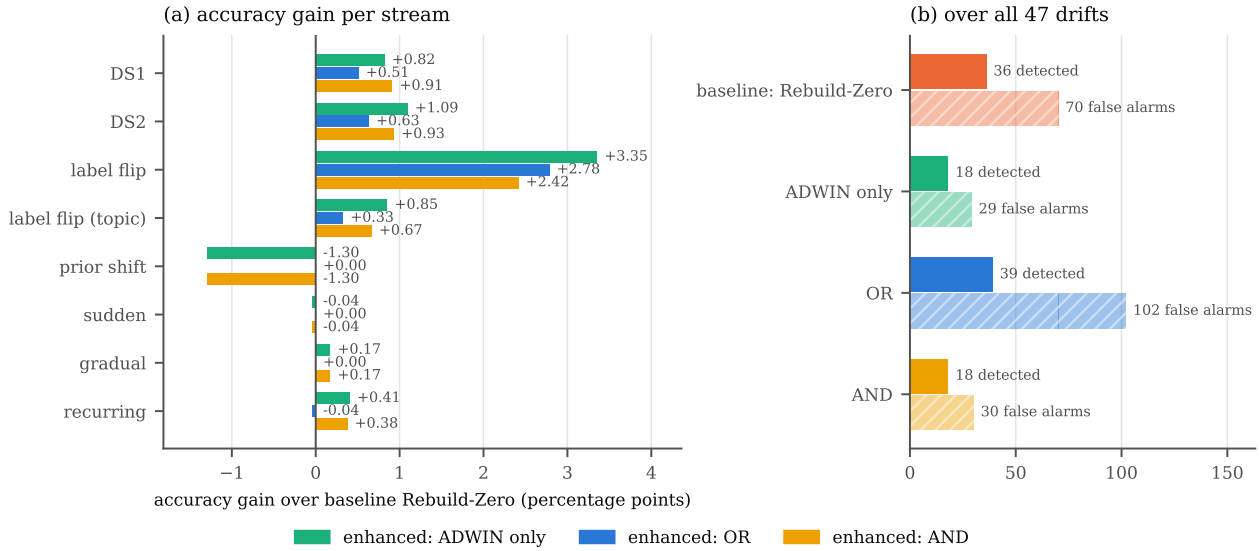


Fig. 5. Fusion modes: (a) accuracy gain over the baseline (Rebuild-Zero) per stream; (b) detected drifts and false alarms over all 47 drifts.

and fading MNBs never recover within the stream; the baseline stays at about 25% until its vocabulary detector fires one window (3,999 instances) later, and misses the flip entirely in one of five seeds. The enhanced model detects

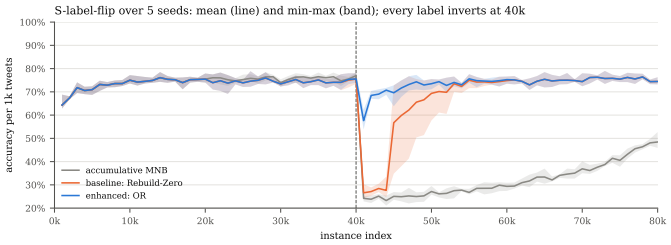


Fig. 6. S-label-flip over 5 seeds: mean accuracy per 1k tweets (lines) and min–max over seeds (bands). All labels invert at 40k.

the flip after 39 instances, dips to about 58%, is back near 70% about 2,000 instances after the flip and reaches 95% of its pre-drift level after 3,183 instances (baseline: 8,204); its accuracy is $73.8 \pm 0.3\%$ against $71.0 \pm 1.0\%$, and the spread over seeds is smaller. When only topic A's labels invert, the vocabulary detector misses the drift in 2 of 5 runs while ADWIN detects all five after about 120 instances (Fig. 4). Accuracy rises only from 63.1% to 63.5–64.0% there, because MNB scores words independently and cannot represent a polarity change limited to one topic: every model is capped at about 64% after that drift.

C. Detection, False Alarms and Recovery

Over all 47 true drifts (Fig. 4, Table II) the OR model detects 39 with a mean delay of 2,290 instances, against 36 and 4,766 for the baseline. On DS1 ADWIN flags the natural change point 704 instances after it, against 19,224 for the vocabulary detector; on DS2 3,124 against 11,012. The price is false alarms: 102 for OR against 70, almost all from the vocabulary channel on DS1 and DS2. Recovery (Fig. 7) is 2.6× faster on the label flip (3,183 vs 8,204 instances). On DS1 the enhanced model never falls below 95% of its pre-change accuracy, while the baseline needs 2,384 instances; on DS2 the enhanced model recovers slightly more slowly (5,560 vs 4,610 instances). For the sudden and gradual topic switches accuracy never falls below the recovery target; on the recurring scenario both models need 2,436 instances, and on all three virtual drifts the models stay within 0.7 points of each other in accuracy.

D. Fusion Modes

Fig. 5 compares the fusion modes. ADWIN alone gives the largest gains on the label flips and DS2 and the fewest false alarms (29), but it ignores pure prior shifts and topic switches because they do not raise the error: on the prior shift it loses 1.3 points, as does AND. OR keeps the vocabulary channel and with it the prior-shift detection: it never falls more than 0.04 points below the baseline on any stream, detects the most drifts and matches or beats the baseline in 28 of 32 stream–seed pairs. We therefore use OR as the main model and report the others as ablations.

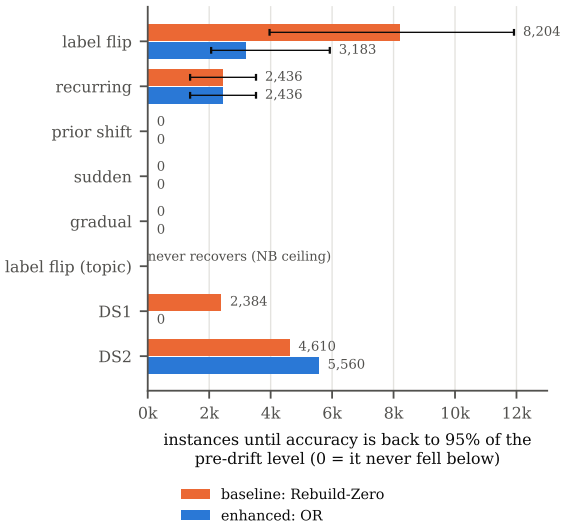


Fig. 7. Recovery time per stream (mean, whiskers: min–max over seeds).

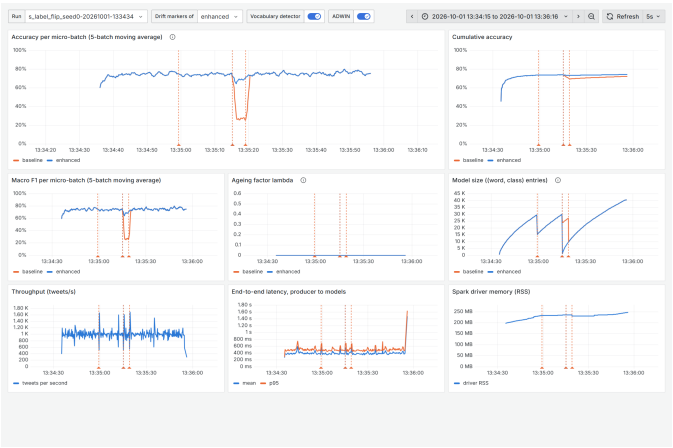


Fig. 8. Grafana during a live S-label-flip run (1,000 tweets/s): baseline (orange) and enhanced (blue) side by side, drift annotations (orange: vocabulary detector, blue: ADWIN).

E. Streaming Performance and Cost

Through Kafka and Spark the two models reproduce the offline accuracies (72% and 74% on the label flip at 1,000 tweets/s), and the dashboard shows the enhanced model's ADWIN marker at the flip while the baseline sits near 25% (Fig. 8). With micro-batches capped at 1,000 tweets the pipeline keeps up with 2,000 tweets/s at a median latency of 0.42 s (p95 0.76 s) and saturates at about 2,840 tweets/s; with a cap of 5,000 it sustains 5,593 tweets/s at a median latency below 1 s. The bottleneck is Spark's fixed cost of about 170 ms per micro-batch, not the models: the baseline needs 22.6 ms and the enhanced model 32.9 ms per 1,000 tweets (Table III); ADWIN itself costs under 1 μs per update. Offline, all adaptive models process 37–46k tweets/s.

TABLE III
Cost of the enhancement. Offline: mean over the grid runs. Spark: model time per 1k tweets in the streaming pipeline (S-label-flip, 1,000 tweets/s); the pipeline adds about 170 ms fixed overhead per micro-batch for both.

Model	Offline tweets/s	Batch p50 / p95 (ms)	Entries	Spark ms per 1k
baseline: Rebuild-Zero	38,303	29 / 55	41,934	22.6
enhanced: OR	37,213	27 / 58	40,088	32.9

VI. Discussion

The results support the motivating hypothesis: monitoring the classifier's error complements monitoring its input. The vocabulary detector is the right signal for changes in $P(X)$ and $P(y)$ (topic and prior shifts); ADWIN on the error is the right signal for changes in $P(y \mid X)$, and it is fast because it updates on every tweet rather than every w tweets. Three further observations matter for practice. First, virtual drift barely affects an MNB classifier, so detecting it is informative but not necessary for accuracy; an error-based trigger avoids needless rebuilds. Second, λ-based strategies depend on the time scale: SlowIncrease-FastReset is the best baseline on DS1 (81.4%) but reaches only 54.5% on the label flip, because the scenarios span about 22 synthetic hours and a larger λ per hour barely ages the model; rebuilding does not depend on the time unit. Third, the cheapest way to handle false alarms is a cheap reaction: rebuilding from recent data costs little accuracy when a false alarm occurs, which is why the OR model still improves accuracy despite more false alarms.

VII. Limitations

Sentiment140 labels are derived from emoticons and noisy (1,685 tweets appear with both labels). The controlled drifts are synthetic, the topics are defined by keywords and the real streams' ground truth contains only designed drift points, so false alarms there are upper bounds. DS2's removal pattern is our own because the paper does not specify it, and our accumulative baseline is stronger than the paper's, which narrows the gaps we measure. The models are sequential, so Spark parallelises only ingestion and tokenisation; the model state lives on the driver and is lost on restart. Finally, the data is from 2009, so vocabulary and usage differ from today's platforms.

VIII. Conclusion and Future Work

We reproduced informed ageing MNB on Sentiment140 and showed that its vocabulary-based detector misses or delays drift that changes meaning without changing words. Adding ADWIN on the prequential error, fused with the vocabulary detector and combined with a rebuild on the post-drift window, detects such drift about 100 times sooner, recovers 2.6 times faster and improves accuracy on both real streams at negligible cost, in an end-to-end Kafka/Spark/Grafana pipeline. Future work includes a classifier that can model context-dependent polarity (to lift the naive Bayes ceiling we observed), drift-specific reactions (e.g. reweighting the class prior on a detected prior shift), keyed state so that Spark can partition the stream across executors, and evaluation on recent, event-driven streams such as the one studied by Bechini et al. [1].

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